

100% Excellent!

Name: _____

These questions assess your understanding of the material from Chapter(s) 20, 21 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. What current flows through a high-voltage transmission line to transmit 60. MW of power at 170 kV?

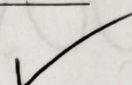
ANSWER: $\frac{60 \times 10^6}{170 \times 10^3} = 352.94 \text{ A}$

$$P = IV$$

$$I = \frac{P}{V}$$

$$I = \frac{60 \times 10^6}{170 \times 10^3}$$

$$I = 352.94$$



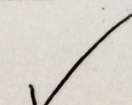
2. Calculate the power dissipated by a 2.8-Ω car headlight when it has a voltage applied to it of 13. V.

ANSWER: $\frac{13^2}{2.8} = 60.36 \text{ W}$

$$P = \frac{V^2}{R}$$

$$P = \frac{13^2}{2.8}$$

$$P = 60.357$$



3. What is the value of the peak voltage for 190-V AC power?

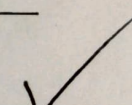
ANSWER: 268.7 V

$$V = \frac{V_0}{\sqrt{2}}$$

$$\sqrt{2} V = V_0$$

$$V_0 = \sqrt{2} (190)$$

$$V_0 = 268.70057$$

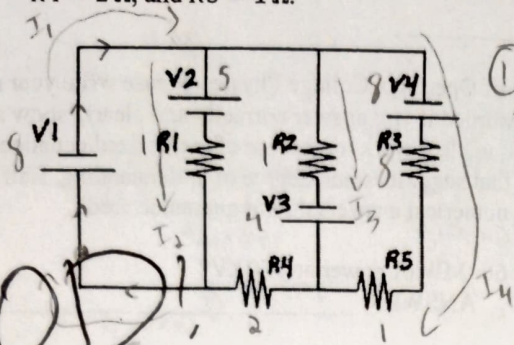


+3

$$I_4 = I_1 - I_2 - I_3$$

$$V = IR \quad V = (1.517)(1) = 1.517$$

4. Solve for the voltage drop across resistor R_2 given $V_1 = 8 \text{ V}$, $V_2 = 5 \text{ V}$, $V_3 = 4 \text{ V}$, $V_4 = 8 \text{ V}$, $R_1 = 1 \Omega$, $R_2 = 1 \Omega$, $R_3 = 8 \Omega$, $R_4 = 2 \Omega$, and $R_5 = 1 \Omega$.



ANSWER: 1.517V

$$\begin{aligned} \textcircled{1} \quad -V_1 - V_2 - R_1 I_2 &= 0 \\ -8 - 5 - I_2 &= 0 \\ -I_2 &= 13 \\ I_2 &= -13 \end{aligned}$$

$$\textcircled{2} \quad -V_1 - R_2 I_3 + V_3 - R_4(I_1 - I_2) = 0$$

$$-8 - I_3 + 4 - 2I_1 + 2I_2 = 0$$

$$-2I_1 = -2I_2 + I_3 + 4 \quad I_1 = -\frac{-2I_2 + I_3 + 4}{2}$$

$$\textcircled{3} \quad -V_1 + V_4 - R_3 I_4 - R_5 I_4 - R_4(I_1 - I_2) = 0$$

$$-8 + 8 - 8I_1 + 8I_2 + 8I_3 - I_1 + I_2 + I_3 - 2I_1 + 2I_2 = 0$$

$$-11I_1 + 11I_2 + 9I_3 = 0$$

$$+ \frac{11(-2(-13) + I_3 + 4)}{2} + 11(-13) + 9I_3 = 0$$

$$\frac{286 + 11I_3 + 44}{2} - 143 + 9I_3 = 0$$

$$330 + 11I_3 - 286 + 18I_3 = 0$$

$$29I_3 = -44 \quad I_3 = -1.517$$

5. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$) and has a resistance of 0.58Ω . If the filament is a cylinder 80.-cm long, what is its diameter?

0.8m

ANSWER: $3.136 \times 10^{-4} \text{ m}$

$$R = \frac{\rho L}{A}$$

$$A = \frac{\rho L}{R}$$

$$A = \frac{(5.6 \times 10^{-8})(0.8)}{0.58}$$

$$A = 7.724 \times 10^{-8}$$

$$A = \pi r^2$$

$$\pi r^2 = 7.724 \times 10^{-8}$$

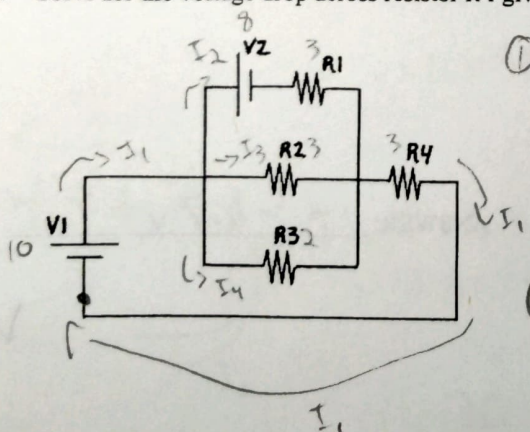
$$r = \sqrt{\frac{7.724 \times 10^{-8}}{\pi}}$$

$$r = 1.568 \times 10^{-4}$$

$$d = 3.136 \times 10^{-4}$$

6. Solve for the voltage drop across resistor R_4 given $V_1 = 10 \text{ V}$, $V_2 = 8 \text{ V}$, $R_1 = 3 \Omega$, $R_2 = 3 \Omega$, $R_3 = 2 \Omega$, $R_4 = 3 \Omega$.

ANSWER: 6V



$$\textcircled{1} \quad V_1 - V_2 - R_1 I_2 - R_4 I_1 = 0$$

$$10 - 8 - 3I_2 - 3I_1 = 0$$

$$-3I_2 = 3I_1 - 2$$

$$I_2 = -\frac{3I_1 - 2}{3}$$

$$\textcircled{2} \quad V_1 - R_2 I_3 - R_4 I_1 = 0$$

$$10 - 3I_3 - 3I_1 = 0$$

$$-3I_3 = 3I_1 - 10$$

$$I_3 = -\frac{3I_1 - 10}{3}$$

$$I_1 = I_2 + I_3 + I_4$$

$$I_1 = -\frac{3I_1 - 2}{3} - \frac{3I_1 - 10}{3} - \frac{3I_1 - 10}{2}$$

$$I_1 = \frac{-6I_1 + 12}{3} - \frac{3I_1 - 10}{2}$$

$$I_1 = \frac{-12I_1 + 24}{6} - \frac{9I_1 - 30}{6}$$

$$I_1 = \frac{-21I_1 + 54}{6}$$

$$6I_1 = -21I_1 + 54$$

$$27I_1 = 54$$

$$I_1 = 2$$

$$\textcircled{3} \quad V_1 - R_3 I_4 - R_4 I_1 = 0$$

$$10 - 2I_4 - 3I_1 = 0$$

$$-2I_4 = 3I_1 - 10$$

$$I_4 = -\frac{3I_1 - 10}{2}$$

$$V = IR$$

$$V = (2)(3) \quad V = 6$$

+3

7. A 12.3-V car battery with internal resistance $0.014\ \Omega$ supplies a starter motor drawing 660 A for 3.9 s. Calculate the electrical energy delivered to the starter during this time.

ANSWER: 7876.44 J

$V = \mathcal{E} - Ir$

note: voltage drop at starter is the terminal voltage

$V = 12.3 - (0.014)(660)$

$V = 3.06$

$E = (660)(3.06)(3.9)$

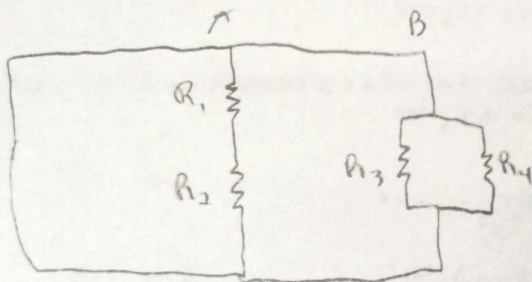
$E = 7876.44\text{ J}$

$\frac{E}{t} = IV$

$E = IVt$

8. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 7.7\ \Omega$ and $R_2 = 1.3\ \Omega$ in series; branch B consists of resistors $R_3 = 5.9\ \Omega$ and $R_4 = 1.8\ \Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.

ANSWER: 1.175 Ω



$R_s = R_1 + R_2 = 9\ \Omega$

$R_p = (R_3^{-1} + R_4^{-1})^{-1} = 1.377\ \Omega$

$R_{pt} = (R_s^{-1} + R_p^{-1})^{-1} = 1.1759\ \Omega$

2.8×10^{-4}

9. Suppose a 0.28-mA current flows through a handheld calculator. How long does it take a charge of $4.5 \times 10^{-5}\text{ C}$ to flow through the calculator?

ANSWER: 0.161 s

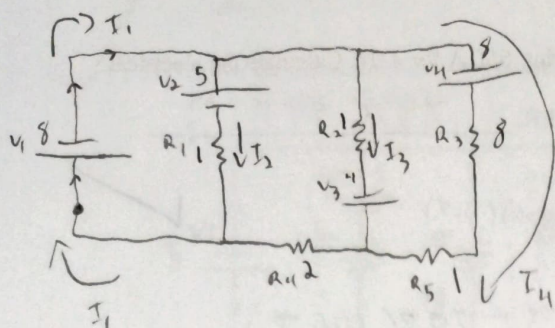
$I = \frac{Q}{t}$

$t = \frac{Q}{I}$

$t = \frac{4.5 \times 10^{-5}}{2.8 \times 10^{-4}}$

$t = 0.1607$

+3



$$I_4 = I_1 - I_2 - I_3$$

$$\textcircled{1} -v_1 - v_2 - R_1 I_1 = 0$$

$$-8 - 5 - I_1 = 0$$

$$-I_1 = 13$$

$$I_1 = -13$$

$$\textcircled{2} -v_1 - R_2 I_3 + v_3 - R_4 (I_1 - I_2) = 0$$

$$-8 - I_3 + 4 - 2I_1 + 2I_2 = 0$$

$$-2I_1 = I_3 - 2I_2 + 4$$

$$I_1 = \frac{-I_3 + 2I_2 - 4}{2}$$

$$\textcircled{3} -v_1 + v_4 - R_3 (I_1 - I_2 - I_3) - R_5 (I_1 - I_2 - I_3) - R_4 (I_1 - I_2) = 0$$

$$-8 + 8 - 8I_1 + 8I_2 + 8I_3 - I_1 + I_2 + I_3 - 2I_1 + 2I_2 = 0$$

$$-11I_1 + 11I_2 + 9I_3 = 0$$

$$-11 \left(\frac{-I_3 + 2(-13) - 4}{2} \right) + 11(-13) + 9I_3 = 0$$

$$\frac{11I_3 + 286 + 44}{2} - 143 + 9I_3 = 0$$

$$11I_3 + 330 - 286 + 18I_3 = 0$$

$$29I_3 = -44$$

$$I_3 = -1.517$$

$$V = IR$$

$$V = (1.517)(1)$$

$$V = 1.517$$

$$I_1 = I_2 + I_3 + I_4$$

$$I_1 = \frac{-3I_1 + 2}{3} + \frac{-3I_1 + 10}{3} + \frac{-3I_1 + 10}{2}$$

$$I_1 = \frac{-6I_1 + 4}{6} + \frac{-6I_1 + 20}{6} + \frac{-9I_1 + 30}{6}$$

$$I_1 = \frac{-21I_1 + 54}{6}$$

$$6I_1 = -21I_1 + 54$$

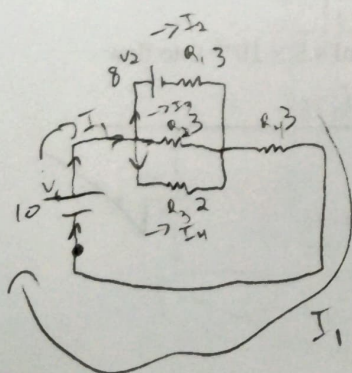
$$27I_1 = 54$$

$$I_1 = 2$$

$$V = IR$$

$$V = 0(3)$$

$$V = 6$$



$$\textcircled{2} v_1 - R_2 I_3 - R_4 I_1 = 0$$

$$10 - 3I_3 - 3I_1 = 0$$

$$-3I_3 = 3I_1 - 10$$

$$I_3 = \frac{-3I_1 + 10}{3}$$

$$\textcircled{3} v_1 - R_3 I_4 - R_4 I_1 = 0$$

$$10 - 2I_4 - 3I_1 = 0$$

$$-2I_4 = 3I_1 - 10$$

$$I_4 = \frac{-3I_1 + 10}{2}$$

PHY202 Exam 3 on chapter(s): 20, 21

1. $I_{RMS} = 353. \text{ A} \therefore P_{ave} = I_{RMS} V_{RMS}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex10)
2. $P = 60.4 \text{ W} \therefore P = \frac{V^2}{R}$
(Chapter 20, Section (4) Electric Power and Energy - cp1e_ch20_ex7)
3. $V_0 = 269. \text{ V} \therefore V_{RMS} = \frac{V_0}{\sqrt{2}}$
(Chapter 20, Section (5) Alternating Current v. Direct Current - cp1e_ch20_ex9)
4. $V = 1.52 \text{ V} \therefore$ Kirchhoff's rules and $V = IR$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)
5. $D = 3.14 \times 10^{-4} \text{ m} \therefore R = \frac{\rho L}{A}$
(Chapter 20, Section (3) Resistance and Resistivity - cp1e_ch20_ex5)
6. $V = 6.00 \text{ V} \therefore$ Kirchhoff's rules and $V = IR$
(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ1)
7. $E = 7.88 \times 10^3 \text{ J} \therefore P = IV = \frac{E}{t}, V = \text{emf} - Ir$
(Chapter 21, Section (2) Electromotive Force: Terminal Voltage - cp1e_ch21_ex4)
8. $R_{eq} = 1.20 \Omega \therefore R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}$
(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex3)
9. $t = 0.161 \text{ s} \therefore I = \frac{\Delta Q}{\Delta t}$
(Chapter 20, Section (1) Current - cp1e_ch20_ex1)